

Master's Thesis

Applications of Reinforcement Learning to Stateful Emulator-Based Greybox Fuzzing

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Abstract

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Chapter 1

Introduction

1.1 Motivation

1.2 Problem Statement and Research Questions

1.3 Contributions

1.4 Outline

Chapter 2

Background

2.1 Fuzzing

2.1.1 Overview and Motivation

Fuzz testing, or *fuzzing* for short, is a widely deployed method for automatic software testing and vulnerability discovery that makes use of a diverse set of tools [28]. Its goal is to find inputs to a Program Under Test (PUT) that trigger a certain behavior, often a crash that might then lead to the discovery of bugs or security vulnerabilities.

The technique was first introduced in the early 1990s as a way of testing the reliability of UNIX utilities [29]. The authors were inspired by spurious characters being introduced to a dial-up line causing crashes of common programs on a remote workstation. This led to the creation of a simple command-line utility *fuzz* that produced random input strings that were then sent to various utilities in the hope of causing a crash in the form of a core dump or a hang. They managed to find inputs that caused between 24% and 33% of the tested utility programs to crash or hang. By analyzing these inputs, bugs could be identified and fixed, thereby contributing to the reliability of the system.

The same approach is still in use today to find security vulnerabilities in input-processing code, especially memory-safety bugs in C/C++. For example, Google’s OSS-Fuzz “aims to make common open source software more secure and stable by combining modern fuzzing techniques with scalable, distributed execution” [2]. It supports a variety of target languages and is therefore widely applicable. OSS-Fuzz has helped to fix over 13,000 security vulnerabilities and 50,000 bugs across 1,000 projects, proving the importance of modern fuzzing techniques for ensuring reliability and security.

2.1.2 Core Terminology

We will introduce some general definitions that mainly follow Manes et al. [28].

1. *Program Under Test (PUT)*: The program under test is an executable software artifact like a command-line utility or library together with an interface that consumes input. In our case it denotes a binary executed through the MOGI emulator (Chapter 5).

2. *Security Policy*: A security policy defines which observable behaviors of the PUT are acceptable. Each execution either satisfies or violates a given security policy. A common security policy deems all executions that result in crashes like segmentation faults to be violations.
3. *Bug Oracle*: A bug oracle is a program that decides whether a given input to the PUT violates a given security policy. A widely used bug oracle might deem an execution a violation if the PUT crashes or hangs.
4. *Fuzzing*: Fuzzing denotes the process of feeding the PUT with inputs sampled from an input space. This input space may include expected and valid inputs as well as invalid, unexpected inputs in the hope of violating a given security policy.
5. *Fuzz Testing*: Fuzz testing denotes the software testing technique that makes use of fuzzing.
6. *Fuzzer*: A fuzzer is a program that performs fuzz testing on a PUT.
7. *Fuzz Campaign*: A fuzz campaign is a specific execution of a fuzzer on a PUT with a specific security policy. Its goal is to find bugs in the PUT that violate this policy.
8. *Test Case*: A test case denotes a concrete input that the fuzzer feeds to the PUT during a fuzzing campaign. The resulting execution is then judged by the bug oracle and the test case recorded in case the bug oracle deems it a violation of the security policy.
9. *Fuzz Configuration*: A fuzz configuration comprises the parameter values that control the fuzzing algorithm.

2.1.3 Taxonomy

Fuzzers can be roughly divided into three classes, defined by the amount of information from the execution of the PUT available to them. [26].

Black-box fuzzing

Black-box fuzzers have no access to the internal behavior of a PUT during execution. They can only make use of the bug oracle to decide whether a given input violates the security policy or not. The core advantage of this approach is the wide applicability since no instrumentation is required. This lack of potential instrumentation overhead also benefits execution throughput. The obvious disadvantage is the lack of information about PUT behavior. This makes it difficult for fuzzers to generate high-coverage inputs.

White-box fuzzing

In contrast to pure black-box fuzzers, white-box fuzzers have full access to the internal logic of the PUT as well as usually any original source code or documentation which can greatly aid the fuzzing process. In practice, this allows an approach known as *symbolic execution*: instead of relying on actual execution traces, a white-box fuzzer can gather symbolic constraints at

branching points and use constraint solving to find inputs that exercise a specific path. While this allows to easily find test cases of interest in theory, this approach scales poorly to large PUTs due to the path explosion problem. A popular white-box fuzzer is SAGE [15].

Grey-box fuzzing

Since full information is often not available but observing certain parts of the internal behavior of the PUT might still be possible, *grey-box* fuzzing represents an important middle-ground between the previous two extremes. By making use of runtime instrumentation information like code coverage, the fuzzer can better judge the behavior of the PUT for a given input and direct its exploration efforts towards promising regions. Limitations are discussed in detail in Section 2.1.6.

Hybrid approaches

There also exists a variety of hybrid approaches combining different fuzzing methodologies. A prominent example is *Driller* that alternates pure greybox fuzzing with *concolic* execution (a combination of concrete and symbolic execution) to overcome hard branch conditions like CNC checks that have historically presented a major obstacle for fuzzers [48]

2.1.4 Input generation

The content of a test case directly controls whether a bug is triggered or not, making it one of the most influential design decisions of a fuzzer [28]. Input generation can be categorized into either generation- or mutation-based.

Generation-based input generation

Generation-based input generation makes use of a model of the input expected by the PUT, often represented through a grammar specification. These models can either be *predefined*, *inferred*, or *encoder-based*.

Predefined models are specified by the user. They are often used by network fuzzers consuming protocol specifications or kernel fuzzers consuming system call interfaces.

Inferred models don't rely on an explicitly specified model but rather infer it through some automatic means. Inference can either take place as a preprocessing step or during a fuzz campaign. Test-Miner [52] searches for literals inside of the PUT and derives suitable inputs. PULSAR[13] infers a network protocol from a set of captured network packages generated by the PUT.

Encoder-based models interpret the PUT as a decoder for inputs generated by an existing encoder.

Mutation-based input generation

Generating purely random inputs is not an effective strategy. In general, a PUT contains checks that reject invalid inputs, making it difficult to reach code paths that violate a given security policy [28]. This motivates the use of a *seed corpus*, a set of well-structured inputs to

the PUT like valid network packages or files. The seeds inside of the seed corpus are then successively mutated with the goal of triggering abnormal behavior. Mutation operators are divided into four sets:

1. *Bit-flipping*: Bit flipping is governed by a *mutation ratio*, the fraction of bits to flip per execution.
2. *Arithmetic mutation*: Arithmetic mutation reinterprets byte sequences as integers and perturbs them.
3. *Block-based mutation*: Block-based mutation inserts, deletes, replaces, permutes or resizes byte blocks, and splices blocks across seeds.
4. *Dictionary-based mutations*: Dictionary-based mutations substitute values with other values of high semantic weight. These usually include numbers like 0, or -1, or format strings for string substitution.

2.1.5 Coverage-Guided Greybox Fuzzing

This work will focus primarily on coverage-guided greybox fuzzing, *CGF* for short. It is the dominant design among modern general-purpose fuzzers, popularized by AFL and its successor AFL++ [11]. We will outline the general approach following Böhme, Pham, and Roychoudhury [4].

cite: original AFL (Zalewski)

Overview

A coverage-guided greybox fuzzer in general uses mutational input generation (Section 2.1.4). For this, it keeps a set of test cases T as part of its runtime state. This set is initialized from a seed corpus S . The fuzzer then selects an input $t \in T$ and derives a new input t' . If the fuzzer deems the execution of the PUT under t' to be interesting, t' is then added to the test case set T .

Algorithm 1 Coverage-guided greybox fuzzing, following Böhme, Pham, and Roychoudhury [4].

Require: Seed corpus S , program under test P

Ensure: Crashing test cases $T_{\times}$, test case set T

```

1:  $T \leftarrow S$ 
2:  $T_{\times} \leftarrow \emptyset$ 
3: if  $T = \emptyset$  then
4:    $T \leftarrow \{\text{empty input}\}$ 
5: end if
6: repeat
7:    $t \leftarrow \text{CHOOSENEXT}(T)$  ▷ seed scheduling
8:    $p \leftarrow \text{ASSIGNENERGY}(t)$  ▷ power schedule
9:   for  $i \leftarrow 1$  to  $p$  do
10:     $t' \leftarrow \text{MUTATEINPUT}(t)$ 
11:    if  $t'$  crashes  $P$  then
12:       $T_{\times} \leftarrow T_{\times} \cup \{t'\}$  ▷ bug oracle
13:    else if  $\text{ISINTERESTING}(t')$  then
14:       $T \leftarrow T \cup \{t'\}$  ▷ coverage check
15:    end if
16:  end for
17: until timeout reached or campaign aborted
18: return  $T_{\times}, T$ 

```

Algorithm 1 states the CGF loop. The fuzzer must solve mainly three problems: selecting an appropriate test case (`CHOOSENEXT`, `ASSIGNENERGY`), applying appropriate mutations (`MUTATEINPUT`), and deciding whether to retain the mutated input or not based on PUT behavior (`ISINTERESTING`). In addition, `ASSIGNENERGY` decides how much of the execution budget the selected test case receives.

Note that two independent decisions are taken after every execution. The bug oracle decides whether the execution is a *violation* and therefore a reportable finding (line 12). In a separate manner, coverage information is used to decide whether the execution is “interesting” and therefore worth keeping as a starting point for further mutation (line 14).

The fuzzing loop can be interpreted as an evolutionary algorithm [42, 28]. The test case set T is a population, the mutation operators of Section 2.1.4 are the genetic operators, and coverage novelty is the fitness function deciding which offspring survive into the next generation. Importantly, the fuzzer doesn’t form an explicit model of what constitutes a good input. It only observes which test cases reach a location no previous test cases reached before. This is a very weak learning signal, and a large part of the research surveyed in Section 2.1.6 and Chapter 4 can be read as an attempt to extract more information from the same executions.

Test case selection

CHOOSENEXT decides which test case $t \in T$ is mutated next, and ASSIGNENERGY decides how many mutants are derived from it before the fuzzer moves on. The two together determine how a finite execution budget is distributed over the test case set. They are commonly referred to as *seed scheduling* and the *power schedule* [28, 4].

An optimal seed and power schedule would choose seeds in such a way as to maximize coverage after they are mutated. This coverage is not known to the fuzzer unless it executes the PUT with a mutated seed, that is, only once the budget has already been spent on that seed. Scheduling is therefore subject to what Manes et al. [28] call the *exploration versus exploitation conflict*: time can either be spent gathering more accurate information about the available choices in order to inform later decisions (*explore*), or on fuzzing the ones currently believed to lead to the most favorable outcome (*exploit*).

Test case mutation

MUTATEINPUT derives a new test case t' from the selected test case t by applying one or more mutation operators to it. Each mutation requires a location in the input to mutate as well as an actual mutation operation.

The main problem is the size of the mutation space. For an input of length L and a mutation set M , the space of all possible mutations has size $O(LM)$. This presents a challenge for long inputs as the mutation space is very large compared to the number of useful mutations. Branch conditions are usually affected by only a few bytes of a given input, making it hard to select appropriate mutations. Some fuzzers try to derive that dependency through data-flow analysis fuzzing [12] and through neural surrogates that approximate the byte-to-coverage map and follow its gradient [42, 43].

Another problem lies in the existence of magic bytes or checksums which are hard to reach by random mutations. For example, a four-byte comparison is satisfied with probability 2^{-32} per attempt, making it very unlikely for a single mutation to overcome it. This motivates a hybrid approach that makes use of concolic execution [48], combining mutations with highly targeted constraint solving.

Choosing appropriate mutations is a scheduling problem that most conventional fuzzers try to solve through a fixed, hand-tuned distribution. This has the major drawback that useful mutations are highly target-dependent. Some fuzzers like MOpt [27] therefore search for a better distribution online using particle swarm optimization, and Rainfuzz [3] learns a heat map over byte positions with a policy-gradient method.

Section 8.5.

Test case retention

ISINTERESTING decides which mutated test cases to keep. Its decision is based on coverage information. Coverage information is recorded by a lightweight program instrumentation framework during execution. The instrumentation can collect coverage at different granularities:

reference
2.1.4:
mutation-
based
input gen-
eration

1. *Basic-block coverage* records which blocks were executed.
2. *Edge or branch coverage* records which transitions (A, B) between blocks were taken and is strictly more informative.
3. *Path coverage* tracks a sequence of block hits $(A, B, \dots)$. While this contains the most amount of useful information, execution paths can grow large and therefore slow down the fuzzer.

Orthogonal to the granularity is the question of how precisely a hit is counted. One bit per coverage element is the cheapest option, but it cannot distinguish how often a block was hit. Most coverage-guided fuzzers therefore keep a hit count per element and quantize it into buckets of logarithmically scaling size, so that a change in hit counts registers as new behavior once a new bucket is entered.

To decide whether or not a new coverage is interesting, it is compared against a global coverage map accumulated over the whole campaign. t' is then deemed interesting if it contains coverage not yet contained in the global coverage map. In practice, coverage maps are implemented as fixed-size arrays indexed by a hash of the coverage element. This enables constant-time coverage comparisons, but may cause aliasing coverage information due to hash collisions.

Instrumentation

Collecting coverage information requires instrumenting the PUT. There are several categories of instrumentation in use [11]:

1. *Compile-time instrumentation* inserts instrumentation during compile time of the PUT. This is very cheap and flexible, but requires access to the source code.
2. *Static binary rewriting* injects the same code into an existing binary and removes the requirement of having source code available, but requires correct disassembly.
3. *Dynamic binary translation* instead instruments blocks as they are translated at runtime, which works on arbitrary binaries and across architectures but costs a considerable slowdown relative to native execution.
4. *Hardware tracing* makes use of built-in hardware instrumentation and tracing mechanisms like Intel Processor Trace [59]. This entails low overhead, but restricts instrumentation to host-native binaries.

AFL++

One famous example of a coverage-guided greybox fuzzer is AFL++ [11]. It is a mutation-based fuzzer that iteratively improves the seed corpus through a series of mutations and selections. Mutations include operations like bit flips, substitutions and splicing. Newly generated inputs are then evaluated and added to the seed corpus for further mutation and

cite: Gan et al. 2018 on coverage map collisions

cite: bel-lard05) for dynamic binary translation

selection if they produce new coverage. The success of a greybox fuzzer is strongly dependent on a high-quality initial corpus and appropriate seed mutation and selection strategies.

AFL++ is a maintained reimplement of the now aged AFL fuzzer [TODO cite original afl]. It contains various refinements and modules implementing the four components of Algorithm 1. Because of this, it often serves as a common baseline in the literature.

Directed greybox fuzzing

Not every campaign wants coverage everywhere. In patch testing, crash reproduction, or the triage of statically flagged suspicious code, the goal is to reach a specific set of program locations. *Directed* greybox fuzzing replaces the novelty objective with a distance between the executed path and those target locations, and uses a power schedule that concentrates energy on the test cases that are closest. Refinements condition on program state rather than on control-flow distance alone [7].

Directed fuzzing is not pursued in this thesis, but it makes an important structural point. The objective in Algorithm 1 is not intrinsic to greybox fuzzing; it is one replaceable component among four, and swapping it yields a different search without changing the loop around it. The formulations developed in Chapter 6 exercise exactly that freedom.

cite:
AFLGo
(Böhme
et al.
2017)
as the
canonical
directed
greybox
fuzzer

2.1.6 Strengths and Limitations of Coverage-Guided Fuzzing

- Strengths (why CGF won):
 - simple, robust, almost no per-target manual effort [28],
 - scales linearly with compute; embarrassingly parallel in principle [62],
 - empirically the most productive bug-finding technique of the last decade [28, 11].
- Limitations — each one motivates “smarter” input generation and frames a design goal of this thesis:
 - *Coverage plateaus*: random mutation practically never satisfies narrow constraints (magic bytes, checksums, length fields), so campaigns stall on frontier branches [48, 42].
 - *Coverage is only a proxy*: maximizing edges is not the same as triggering bugs; state- and data-dependent bugs need specific values in specific contexts, not just new edges [7].
 - *Blind mutation wastes the execution budget*: most mutations are irrelevant to control flow; which bytes and operators matter is input- and target-dependent [27, 3].
 - *Feedback aliasing*: bitmap hash collisions and coarse hit-count bucketing hide genuinely new behavior .
 - *Statelessness*: the classic loop assumes one self-contained input per fresh execution — breaks down for stateful/reactive targets (Section 2.1.7).

cite: Col-
LAFL

- *Evaluation is tricky*: campaigns are highly stochastic; sound comparisons need many trials and statistical testing (also needed for Chapter 11).

- Research responses (survey briefly, as context for the RL approach):
 - data-flow/taint guidance to identify influential input bytes (GreyOne) [12],
 - neural surrogates: smooth the coverage function and follow gradients (NEUZZ, MTFuzz) [42, 43], deep-learning path prediction (NeuFuzz) [56],
 - hybrid concolic execution (Driller) [48],
 - machine learning across all loop stages — seed selection, mutation, scheduling — surveyed in [55, 36].

cite:
Klees et al. 2018, “Evaluating Fuzz Testing”

2.1.7 Stateful and Protocol Fuzzing

- Many high-value targets are reactive servers speaking a protocol (network services, industrial control systems, embedded devices): behavior depends on *session state* accumulated over a message exchange [1, 5].
- Consequences for the classic single-input model:
 - a test case is a *sequence* of messages, not one blob; ordering and protocol context matter,
 - deep states are reachable only through valid message prefixes; mutating a late message is useless if an early one breaks the session,
 - the same code can behave differently in different states, so pure edge coverage under-represents state coverage [5],
 - state is invisible: the fuzzer must infer or track it (response codes, annotated state variables) .
 - state-awareness matters in adjacent domains too, e.g. black-box web scanning [9].
- Extra decision problem on top of the CGF loop: *which state to fuzz next* — an exploration/exploitation trade-off, already modeled as a multi-armed bandit in prior work [5]; direct bridge to Section 2.2.
- Practical complications for encrypted/checksummed transports (e.g. QUIC): the fuzzer must sit inside the crypto boundary to mutate meaningfully [1].
- Position of this thesis: protocol-level actions against a stateful target are exactly the action space handed to the RL agent (Chapter 7).

cite:
AFLNet (Pham et al. 2020) and SGFuzz (Ba et al. 2022) — the standard stateful greybox fuzzers; both are baselines the thesis should discuss

2.1.8 Emulator-Based Fuzzing

- Motivation from embedded/firmware targets: fuzzing on the device is slow and, worse, faults are often *silent* — without MMU protection or sanitizers, memory corruption frequently does not crash (“what you corrupt is not what you crash”) [34].

- Rehosting the target in an emulator restores the missing capabilities [34] :
 - full observability: coverage and memory-access instrumentation without source access,
 - determinism and replayability of executions,
 - fine-grained fault detection via emulator-level checks,
 - *snapshots*: capture full machine state once, restore cheaply before every test.
- Snapshot restore gives an exact, cheap *reset* of the target to a known state — for stateful targets this is what makes per-episode experimentation tractable, and it is precisely the environment-reset primitive that reinforcement learning assumes (Section 2.2); realized here by the MOGI framework (Chapter 5).
- Cost: emulation overhead reduces raw executions per second versus native execution — sharpens the need to make every execution count (sample efficiency, Chapter 3).

cite:
QEMU
(Bellard
2005);
Unicorn
engine;
optionally
a rehost-
ing sur-
vey (e.g.
Wright
et al.
2021 or
Fasano et
al. 2021)

2.2 Reinforcement Learning

2.2.1 Agent, Environment, and Markov Decision Processes

Reinforcement learning (RL) is an extremely general framework for approaching problems in which an agent strives to maximize a reward signal through a selection of actions in certain states [50]. These kinds of problems can be modeled as finite Markov decision processes (MDPs). Formally, a Markov decision process is a 3-tuple $MDP = (S, A, p)$ where

- S is the set of states,
- A is the set of actions where $\mathcal{A}_s \subseteq A$ are actions available in state $s \in S$,
- $p(s', r|s, a) = \Pr\{S_t = s', R_t = r | S_{t-1} = s, A_{t-1} = a\}$ describes the system dynamics, i.e., the probability of encountering a next state s' and reward r at time step t when taking action a in state s at time step $t - 1$. This satisfies the *Markov property*: p depends only on the current state and not the full history.

An RL agent's goal is to learn a policy $\pi : S \times A \rightarrow [0, 1]$ mapping each state s to probabilities of taking each possible action a to maximize the cumulative reward.

- Distinguish from supervised learning: no labeled correct actions, only evaluative (and possibly delayed) feedback; trial-and-error credit assignment [50]; idea traces back to early AI [30]; add [35] as the standard MDP reference next to the excerpt.
- Extend the excerpt's formalism [50]:
 - Markov property: transition depends only on current state and action,
 - episodic vs. continuing tasks; discount factor γ and discounted return $G_t = \sum_k \gamma^k r_{t+k}$,

- state value V^π , action value Q^π ; Bellman (optimality) equations; optimal policy π^* .
- Partial observability: agent often sees an *observation*, not the full state (POMDP setting) [50] — relevant later: a coverage vector is a very partial view of the emulated program’s true state (Section 7.3).

2.2.2 Exploration versus Exploitation

- Fundamental dilemma: exploit the best-known action or explore to find better ones [50].
- Minimal setting: multi-armed bandit (single state); ε -greedy and optimism-in-the-face-of-uncertainty / UCB-style strategies [50, 25].
- Contextual bandit as intermediate step between bandit and full MDP — mirrors the ladder of evaluation targets in Chapter 10 (Sections 10.2.1 to 10.2.3).
- Foreshadow: fuzzing has exactly this structure — keep hammering a productive protocol state vs. probe untested ones [5].

2.2.3 Algorithm Families

Algorithms for finding an optimal π^* can be divided into three classes: value-based, policy-based and actor-critic.

Value-based methods try to learn a value function $Q : \mathcal{S} \rightarrow \mathcal{R}$ mapping states to their expected cumulative reward so that in each step, the action that maximizes the expected reward can be taken. These include well-known methods such as Q-learning, TD(λ) or SARSA.

Policy-based methods try to directly construct an optimal policy π^* without the use of a value function. They have various advantages. For example, they can easily handle continuous action spaces and ensure stochasticity which is important for state space exploration. A well-known classical policy-based algorithm is REINFORCE.

Actor-critic methods synthesize value- and policy-based methods by including information about a state’s value in the policy learning process.

Since classical, tabular methods don’t work well with large state spaces, state-of-the-art methods make use modern-day deep learning advancements. Examples include the family of Deep-Q Network (DQN) algorithms [31, 22] approximating the Q function by neural networks, policy-based gradient-descent methods like Proximal Policy Optimization (PPO) [39] and actor-critic methods like A3C [32]. Impressive results range from the master-level Go program AlphaGo [44] and its more generalized successors AlphaZero [45] and MuZero [38] to Dreamerv3 [19] that is able to learn how to acquire diamonds in the popular video game *Minecraft* without any human input or expert knowledge.

- Foundational citations to attach to the excerpt’s method names: temporal-difference learning [49]; Q-learning .
- Further DQN refinements between DQN and Rainbow: double DQN [20], dueling networks [57], prioritized experience replay [37] (PER returns in EfficientZero, Chapter 3).

cite:
Williams
1992,
“Simple
statistical
gradient-
following
algo-
rithms for
connec-
tionist
reinforce-
ment
learn-
ing” (was
williams1992
in the
proposal)

cite:
Watkins

- Further policy-gradient/actor-critic methods: TRPO [40], SAC [17].
- Broad algorithm surveys [14].
- Practical caveats of deep RL: high variance across seeds, hyperparameter sensitivity, reproducibility concerns [21] — informs the evaluation methodology in Chapter 11.

2.2.4 Model-Free versus Model-Based RL

- Model-free: learn policy/values directly from transitions; conceptually simple, but notoriously sample-hungry — millions to billions of environment steps [31, 33].
- Model-based: additionally learn a *dynamics model* of the environment and use it for planning or imagined training — the main lever for sample efficiency [33, 24].
- World-model line: learn compressed latent dynamics, train the policy inside the model [16, 18, 19].
- Planning-with-search line:
 - Monte-Carlo tree search (MCTS) [8], UCT: bandit-based selection inside the tree [25], modern review [51],
 - AlphaGo: MCTS + learned value/policy networks, with known rules [44]; AlphaGo Zero: self-play only [46]; AlphaZero: generalized across games [47],
 - MuZero: replaces the known simulator with a *learned latent model* — planning without knowing the rules [38],
 - EfficientZero: MuZero made sample-efficient (self-supervised consistency, value prefix, model-based off-policy correction); strong Atari performance from only 100k steps (≈ 2 hours) of experience [58] — treated in depth in Chapter 3.

2.2.5 Sample Efficiency

- Define: performance achieved per environment interaction; the binding constraint whenever a step is expensive [33, 58].
- In this thesis each environment step is a program execution under emulation (Section 2.1.8) — orders of magnitude slower than a simulator frame, so the fuzzing use case sits squarely in the low-sample regime that EfficientZero targets [58].
- This is the central argument for choosing a model-based, planning-heavy algorithm over the model-free methods used by most prior RL fuzzers (Chapter 4).

Chapter 3

EfficientZero

Chapter 4

Related Work

4.1 Fuzzing as a Reinforcement Learning Problem

- Why the fit is natural: fuzzing is sequential decision making under uncertainty over an enormous search space, with a cheap scalar progress signal (new coverage) — exactly the RL interface of Section 2.2.1; first formalized as an MDP by Böttinger et al. [6].
- The generic correspondence (present as a table; concrete instantiation for this thesis in Chapter 6):
 - *environment* = the target program (plus harness/emulator),
 - *state/observation* = execution feedback: coverage information, program or protocol state, current input [6, 5],
 - *action* = a fuzzing decision: mutation operator/position, seed choice, protocol message [6, 3, 5],
 - *reward* = newly discovered coverage (or crashes, execution-time properties) [6, 53],
 - *episode* = one test case or one protocol session, ended by a target reset.

4.2 Prior Work

- Organize the survey by *which decision in the fuzzing loop* is learned (following the taxonomy of ML-for-fuzzing surveys [55, 36]):
- **Where/how to mutate:**
 - deep Q-learning over mutation actions on input substrings [6],
 - Rainfuzz: PPO-trained heat maps scoring byte positions by expected usefulness, run collaboratively with AFL++ [3],
 - (non-RL baseline for the same problem: MOpt’s particle-swarm mutation-operator scheduling [27]).
- **Which seed/task to schedule:**
 - multi-armed-bandit / hierarchical seed scheduling for greybox fuzzing [54],

- SyzVegas: MAB-based task and seed selection inside the Syzkaller kernel fuzzer [53],
- Alphuzz: MCTS over the seed-mutation tree [60] — closest prior use of tree search, but over seeds, not program interaction.
- **Which grammar construct / input token to use:** BanditFuzz for SMT solvers [41]; BertRLFuzzer combines a language model with RL-chosen attack mutations [23].
- **Which protocol state to fuzz** (stateful setting, closest to this thesis): stochastic and adversarial MABs for state selection in industrial-protocol fuzzing [5].
- Adjacent security applications showing the same pattern: curiosity-driven web testing [61], Q-learning agents for SQL injection [10], DQN-based autonomous penetration testing [63].

4.3 Recurring Difficulties

- *Reward design:* coverage rewards are sparse and non-stationary — each edge pays out only once, so the reward distribution decays as the campaign progresses; motivates the adversarial-bandit view [5] and reward-model care in [53].
- *Observation design:* what the agent sees (raw bytes, coverage bitmap, protocol responses) is a heavily aliased, partial view of true program state (cf. Section 2.2.1) [6].
- *Throughput mismatch:* a classic fuzzer executes thousands of inputs per second; NN inference per decision must earn back its latency — one reason prior work often keeps the RL component advisory/parallel to a conventional fuzzer [3].
- *Sample cost:* model-free learners need more interactions than a fuzzing campaign can afford, especially under emulation (Section 2.2.5).
- *Mixed empirical results:* learned components do not consistently beat well-tuned static heuristics — e.g. AFLNet outperformed both bandit models in [5]; deep-RL evaluation pitfalls compound this [21].
- Honest framing for the thesis: these difficulties define the burden of proof for the approach evaluated in Chapter 12.

4.4 The Gap Addressed by This Thesis

- Common shape of prior work: a *model-free* learner (Q-learning, PPO, bandits) tunes *one heuristic knob* of an otherwise conventional random fuzzer [6, 3, 54, 53, 41].
- Unexplored combination — the subject of this thesis:
 - a *model-based, planning* algorithm (EfficientZero: learned latent dynamics model + MCTS [58, 38]) as the decision maker,

- acting *directly* at the protocol level against a *stateful* target — the agent drives the session, rather than advising a random mutator (Chapter 7),
 - on targets rehosted in an emulator (MOGI, Chapter 5), whose snapshots provide the exact, cheap episode resets and basic-block coverage observations the MDP framing requires (Section 2.1.8),
 - rationale: the learned model lets planning happen in latent space, amortizing expensive real executions — directly attacking the sample-cost and throughput objections above (Section 2.2.5).
- Research questions — promoted to Section 1.2 as numbered RQs; keep this list as the motivating form and answer them in Chapter 12:
 - Can EfficientZero learn useful policies on fuzzing-shaped reward landscapes (sparse, decaying, heavily aliased observations)?
 - How does the approach scale from toy targets (Chapter 10) up to a real protocol implementation (open62541, Section 10.3.2)?
 - Where does planning with a learned model help or fail compared to classic greybox baselines (Sections 14.1 and 14.2)?

Chapter 5

The MOGI Platform

5.1 Overview

5.2 Emulation and the Guest Model

5.3 Coverage Instrumentation

5.4 Snapshots and the Reset Primitive

5.5 Determinism

5.6 Throughput

Chapter 6

Fuzzing as a Markov Decision Process

6.1 The Environment Interface

6.2 Constraints Imposed by the Learner

6.3 Design Axes

6.4 The Two Formulations

Chapter 7

Input-Level Action Spaces

7.1 Overview

7.2 Action Space

7.3 Observation

7.4 Reward and Episode Structure

7.5 Determinism and the Planning Contract

7.6 Relation to Prior Work

Chapter 8

Corpus-Level Action Spaces

8.1 Overview

8.2 Action Space

8.3 Observation

8.4 Reward and Episode Structure

8.5 Determinism and the Planning Contract

8.6 Relation to Prior Work

Chapter 9

Implementation

9.1 An EfficientZero Implementation in Rust

9.2 The Environment Abstraction

9.3 The Generic Corpus Module

9.4 Experiment and Diagnostic Infrastructure

9.5 Reproducibility

Chapter 10

Targets

10.1 Overview

10.2 Diagnostic Microbenchmarks

10.2.1 Bandit

10.2.2 Contextual Bandit

10.2.3 Sequence

10.2.4 Branches

10.3 Realistic Targets

10.3.1 cJSON

10.3.2 open62541

Chapter 11

Experimental Methodology

11.1 Experimental Setup

11.2 Metrics

11.3 Baselines

11.4 Currencies: Executions versus Wall-Clock

11.5 Multi-Seed Protocol

11.6 Hyperparameter Justification

11.7 Exploratory and Confirmatory Results

Chapter 12

Results

12.1 Input-Level Action Spaces

12.2 Corpus-Level Action Spaces

12.3 Head-to-Head: cJSON under Both Formulations

12.4 Ablations

Chapter 13

Failure Modes and Diagnosis

13.1 Overview

13.2 Policy Collapse and the Echo Chamber

13.3 Dead-Branch Value Fantasy

13.4 Value-Prefix and Support Pathologies

13.5 Reward Hacking

13.6 Diagnosing a Collapsed Run

Chapter 14

Discussion

14.1 Where the Approach Helps

14.2 Costs and Limitations

14.3 Threats to Validity

Chapter 15

Conclusion

Chapter 16

Future Work

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Appendix A

Frozen Configuration

Appendix B

Action and Observation Encodings

Appendix C

Additional Results